Introduction 2

Topics

- **Objects**
  - types of objects
  - creating/naming an object
  - data types
- **Operators**
  - mathematical
  - relational
  - logical

Objects

Objects

Everything we can create in R is called an object (e.g., numbers, vectors, matrices, functions).

```r
number <- 4
number
```

```
[1] 4
```

```r
my_vector <- c(1,2,3,4)
my_vector
```

```
[1] 1 2 3 4
```

```r
my_matrix <- matrix(nrow = 2, ncol = 2, data = my_vector)
my_matrix
```

```
     [,1] [,2]
[1,]    1    3
[2,]    2    4
```

Functions

Everything we do in R is calling **functions** on objects. Functions allow us to **create** and **modify** objects.

```
1 my_vector
```

```
[1] 1 2 3 4
```

```
1 mean(x = my_vector)
```

```
[1] 2.5
```

```
1 my_vector_mean <- mean(my_vector)
2 my_vector_mean
```

```
[1] 2.5
```

Creating/naming objects

Objects are created with the assignment arrow **<-** or **=**

```
1 x1 <- 3 # name <- object
2 x1
```

```
[1] 3
```

```
1 x2 = 3 # name = object
2 x2
```

```
[1] 3
```

Rules for naming objects

- A name **must start with a letter** and can contain letters, numbers, underscores (_), or dots (.).
- It may also start with a dot (.), but in that case it cannot be followed by a number.

```
1 .3 <- 3
```

```
Error in `0.3 <- 3`:
! invalid (do_set) left-hand side to assignment
```

```
1 .x <- 3
```

- It must **not** contain **special characters** such as #, &, $, ?, etc.

- It must not be a reserved word, i.e. one of the words used by R with a special meaning (?reserved).

```
function <- 3
```

```
Error in parse(text = input): <text>:1:10: unexpected
assignment
1: function <-
              ^
```

```
TRUE <- 3
```

```
Error in `TRUE <- 3`:
! invalid (do_set) left-hand side to assignment
```

```
if <- 3
```

```
Error in parse(text = input): <text>:1:4: unexpected
assignment
1: if <-
       ^
```

Some names are not forbidden but are discouraged

```
1 T
```

```
[1] TRUE
```

```
1 T == TRUE
```

```
[1] TRUE
```

```
1 T <- FALSE;
2
3 T == TRUE  #?????
```

```
[1] FALSE
```

```
1 T == FALSE #?????
```

```
[1] TRUE
```

Using “*.*” in object names (for example, “my.data”) is fine in **R** but is not allowed in **Python**, where “*.*” is part of the language syntax.

Across languages, the *naming conventions* for longer, multi-word variable names favour `snake_case` (for example, “**my_data**”) or `camelCase` (for example, “**myData**”), and **abbreviations** where appropriate (for example, “**unipdData**” rather than “university_of_padova_dataset”).

R is case-sensitive

```r
Name <- "Margherita"
name <- "margherita"
Name == name
```

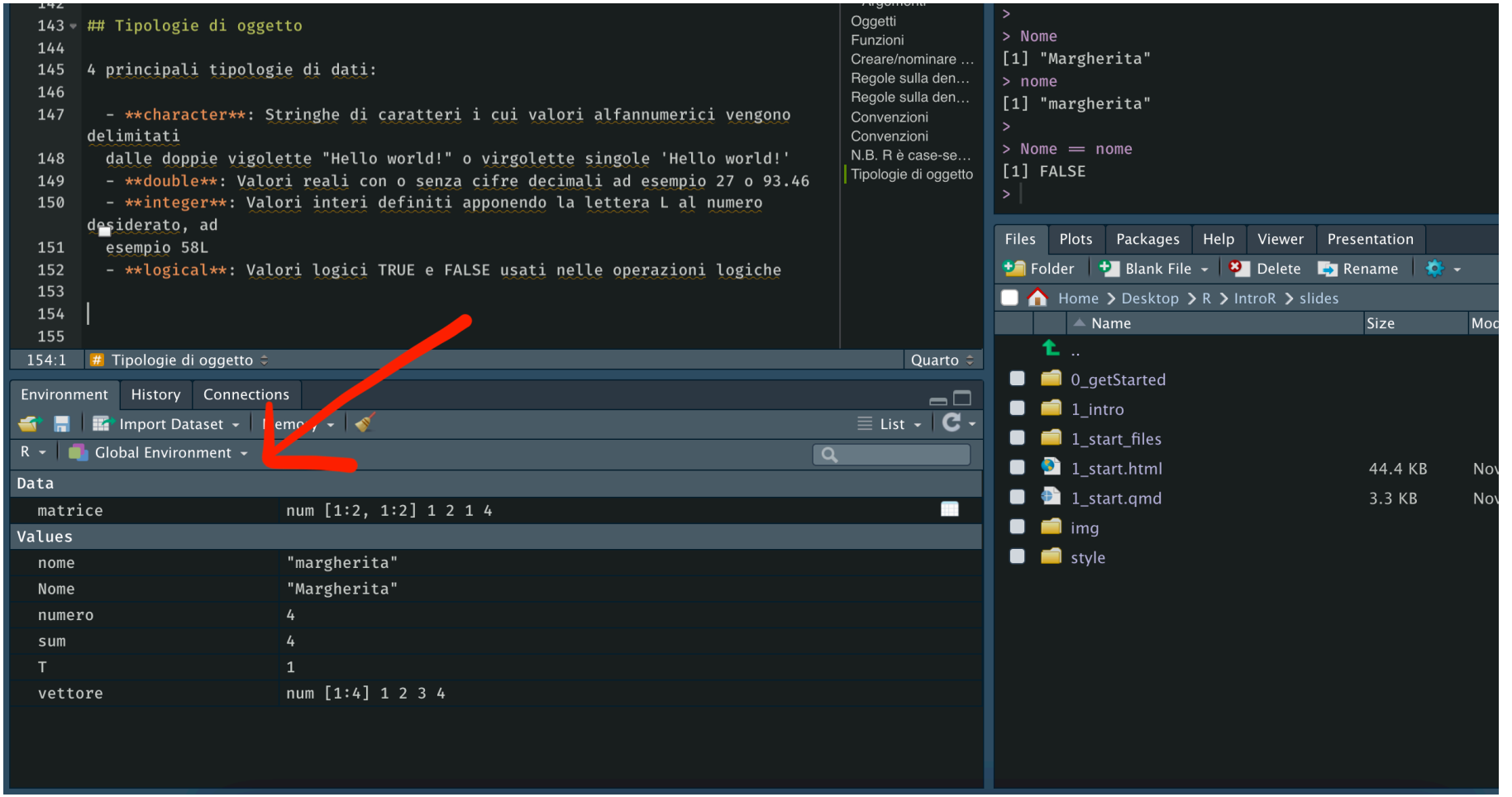
```
[1] FALSE
```

```r
Name <- "margherita"
name <- "margherita"
Name == name
```

```
[1] TRUE
```

Where are objects stored?

By default, objects are created in the **global environment**, which can be accessed with `ls()` or viewed in R Studio together with some additional information:



Removing objects

We can remove an object from our environment with the `rm("objectname")` command.

We can also completely clear/empty our environment with the `rm(list = ls())` command.

Data types

Data types

- `character`: Character strings whose alphanumeric values are delimited by double quotes “Hello world!” or single quotes ‘Hello world!’
- `numbers`:
  - `double`: Real values with or without decimal digits, for example 27 or 93.46
  - `integer`: Integer values defined by appending the letter L to the number, for example 58L

Mathematical Operators

Function	What does it do?	Example	Result
`+`	addition	`5.4 + 6.1`	`11.5`
`-`	subtraction	`9 - 4.3`	`4.7`
`*`	multiplication	`7 * 1.4`	`9.8`
`/`	division	`9/3`	`3`
`%%`	remainder	`9%%2`	`1`
`^`	power	`15 ^ 2`	`225`

Mathematical Operators

Function	What does it do?	Example	Result
`abs`	absolute value	`abs(-8)`	`8`
`sqrt`	square root	`sqrt(225)`	`15`
`exp`	exponential function	`exp(0)`	`1`
`log`	natural logarithm	`log(1)`	`0`
`round`	rounding, integer	`round(1.738)`	`2`
`round`	rounding	`round(1.738, 2)`	`1.74`

Mathematical Operations

The order of operations in R follows the rules of mathematics.

Examples

```r
# Without parentheses
1 + 2 * 3
```

```
[1] 7
```

```r
# With parentheses
(1 + 2) * 3
```

```
[1] 9
```

Relational Operators

In R we can evaluate whether a given relation is true or false. R will evaluate the statement and return `TRUE` if the statement is true or `FALSE` if the statement is false.

Relational Operators

Function	Name	Example	Result
`==`	equal	`30 == 30`	`TRUE`
`!=`	not equal	`30 != 30`	`FALSE`
`>/>=`	greater/or equal	`30 > 10`	`TRUE`
		`30 >= 10`	`TRUE`
`</<=`	less/or equal	`30 < 10`	`FALSE`
		`10 <= 10`	`TRUE`
`%in%`	inclusion	`10%in%c(1,2,10)`	`TRUE`

It doesn't only work with numbers!

```
1 Name <- "Margherita"
2
3 name <- "margherita"
4
5 Name == name
```

```
[1] FALSE
```

NB. Remember that **=** is different from **==**

Logical Operators

In R we can combine several relations to evaluate a given statement.

```r
x <- 30 # Assign the value 30 to x.
```

Function	Name	Example	Result
`&`	Conjunction (AND)	`x>25 & x<60`	`TRUE`
`|`	Inclusive disjunction (OR)	`x>25 | x>60`	`TRUE`
`!`	Negation (NOT)	`!(x<18)`	`TRUE`

Logical Operators

```r
x <- 30 # Assign the value 30 to x.
x
```

```
[1] 30
```

```r
x>25 & x>60
```

```
[1] FALSE
```

```r
x>25 | x>60
```

```
[1] TRUE
```

```r
!(x>18)
```

```
[1] FALSE
```

TRUE is equivalent to 1 and FALSE to 0

```
1 TRUE == 1
```

```
[1] TRUE
```

```
1 TRUE == 2
```

```
[1] FALSE
```

```
1 FALSE == 0
```

```
[1] TRUE
```

```
1 FALSE == 1
```

```
[1] FALSE
```

TRUE is equivalent to 1 and FALSE to 0

I create an example vector containing the words home (repeated 3 times) and shop (repeated 2 times):

```
example <- c("home", "home", "home","shop", "shop") #?rep
example
```

```
[1] "home" "home" "home" "shop" "shop"
```

```
# Which words in `example` are equal to home?
example == "home"
```

```
[1]  TRUE  TRUE  TRUE FALSE FALSE
```

```
sum(example == "home") # why?
```

```
[1] 3
```

Order of Evaluation

When evaluating whether statements are true, R performs the operations in the following order:

1. Mathematical operators (e.g., ^, *, /, +, -, etc.)
2. Relational operators (e.g., <, >, <=, >=, ==, !=)
3. Logical operators (e.g., !, &, |)

If in doubt about the order in which operations are executed, the best thing to do is to use round brackets ( ) to remove any possible ambiguity.

Let’s practice!

Open this link and keep it open: https://etherpad.wikimedia.org/p/arca-corsoR

